MODULE 1 — USER AUTHENTICATION

The User Authentication Module provides secure and personalized access to the Disaster Help Center application. It ensures that users can safely access emergency services, manage their profile information, view SOS requests, and use disaster assistance features. The module acts as the primary security layer of the application and helps protect user information from unauthorized access.

Users can create an account by providing required details such as name, mobile number or email address, password, and basic profile information. During registration, the system validates the entered information and creates a unique user account using Firebase Authentication. Existing users can securely log in using their registered credentials.

The module also supports session management and profile-based access. For emergency situations, anonymous authentication can be used so that a user can send an SOS alert without completing the normal login process. This allows emergency assistance to be requested quickly when immediate action is required.

Main Features:

• User registration and account creation

• Secure login and logout

• Credential validation

• Firebase Authentication

• Email or OTP verification

• Password reset and account recovery

• Session management

• Anonymous authentication for emergency SOS

• User profile management

• Secure access to personalized emergency data

Working of the Module:

The user enters their login credentials through the Flutter application. The credentials are securely processed using **Firebase Authentication**. After successful authentication, the application loads the user's profile and provides access to the relevant emergency services and personalized information.

For emergency situations, the application can create an anonymous authentication session when the user is not logged in. This allows the SOS request to be securely associated with the user and stored in Firebase Firestore without requiring normal account login.

MODULE 2 — HOME / DASHBOARD

The Home or Dashboard Module is the central screen of the Disaster Help Center application. It provides users with quick access to important emergency services and displays essential information in a simple and understandable format. The dashboard allows users to quickly access SOS assistance, location services, emergency contacts, volunteer support, disaster information, and other important features without navigating through multiple screens.

The dashboard provides a summary of the user's current location, emergency status, recent SOS activity, available assistance, and important alerts. It can also provide quick access to features such as SOS Emergency, Map & Location, Volunteer Assistance, Emergency Contacts, Weather Information, and Profile. The design focuses on providing essential emergency functions quickly, especially during critical situations.

Personalized information can be displayed according to the user's profile and current emergency status. For example, the dashboard can show the user's current location, active SOS request, volunteer response status, or important disaster alerts. This makes the application useful as a centralized emergency assistance platform.

Main Features:

• Emergency status overview

• Quick SOS emergency access

• Current location display

• Map and location access

• Emergency contact access

• Volunteer assistance status

• Disaster and emergency alerts

• Weather information

• Recent SOS activity

• Quick access to application modules

• Personalized user information

Working of the Module:

After authentication, the Flutter application loads the user's relevant information and emergency data from Firebase. The dashboard retrieves the user's profile, current location, SOS status, volunteer response, and other available emergency information from the required Firebase services.

The retrieved information is displayed on the Home screen through interactive cards and quick-access buttons. When the user presses the SOS option, the application moves to the emergency assistance workflow, where the current GPS location is obtained and the SOS request is processed according to the available network connection.

MODULE 3 — SOS EMERGENCY & ALERT

The SOS Emergency and Alert Module is the core emergency-response component of the Disaster Help Center application. It allows users to quickly send an emergency request when they are affected by a disaster or require immediate assistance. The module captures the user's current location and emergency details and sends the SOS request to the appropriate emergency support system.

The module is designed to work in both online and offline conditions. When an internet connection is available, the SOS request is stored in Firebase Firestore and can be accessed by volunteers for further assistance. When the internet connection is unavailable, the application can use SMS to send the emergency information, including the user's location, to a configured emergency contact.

Main Features:

• One-tap SOS emergency button

• Automatic GPS location detection

• Online SOS request submission

• Offline SOS support through SMS

• Emergency type selection

• Emergency description

• Number of people affected

• Current address and coordinates

• SOS status tracking

• Firebase Firestore integration

• Volunteer response and assignment

• Real-time emergency status updates

Working of the Module:

When the user activates the SOS button, the Flutter application obtains the current GPS coordinates using the location service. The application also retrieves the corresponding address and collects the required emergency details.

The application then checks the network connection. If the user is online, the SOS information is stored in Firebase Firestore with the user's location, emergency details, and current status. Volunteers can access pending SOS requests and respond to them.

If the user is offline, the application uses SMS to send the emergency information and GPS coordinates to a configured emergency contact. Once an SOS request is created, its status can be updated as Pending, Accepted, Volunteer Arriving, or Resolved, allowing the user to track the progress of the emergency response.

MODULE 4 — LOCATION & MAP SERVICES

The Location and Map Services Module provides real-time location information and map-based visualization during emergency situations. It helps users and volunteers understand the exact location of an emergency request.

The module uses GPS to obtain latitude and longitude and uses geocoding services to convert coordinates into a readable address. Google Maps can be used to display the user's location and volunteer location.

Main Features:

• GPS location detection
• Latitude and longitude retrieval
• Address identification
• Google Maps integration
• User location marker
• Volunteer location marker
• Real-time location updates
• Emergency location sharing
• Map-based emergency tracking
• Route visualization

Working of the Module:

The Flutter application requests the device's current GPS location using the location service. The coordinates are converted into an address using geocoding services.

The location is displayed on Google Maps. When a volunteer accepts an SOS request, the volunteer's location can also be retrieved and displayed together with the user's emergency location.

MODULE 5 — VOLUNTEER MANAGEMENT

The Volunteer Management Module connects emergency users with available volunteers who can provide assistance during disaster situations. Volunteers can view active SOS requests and respond according to the emergency information provided.

The module allows volunteers to view emergency details such as user name, emergency type, location, description, and number of people affected.

Main Features:

• Volunteer registration
• Volunteer profile management
• Active SOS request listing
• Emergency details display
• User location viewing
• Accept SOS request
• Reject SOS request
• Volunteer assignment
• Volunteer availability status
• Volunteer location tracking

Working of the Module:

The volunteer application retrieves pending SOS requests from Firebase Firestore. The volunteer can view the emergency information and choose to accept or reject the request.

When a volunteer accepts an SOS request, the system stores the volunteer ID and location with the SOS record. The status is then updated so that the user can see that assistance has been assigned.

MODULE 6 — EMERGENCY CONTACT & OFFLINE SMS

The Emergency Contact and Offline SMS Module provides an alternative communication method when internet connectivity is unavailable. This is important because disaster situations may affect mobile data or internet services.

Users can configure emergency contacts who can receive SOS information through SMS. The message can contain the user's name, emergency type, GPS coordinates, address, and a map link.

Main Features:

• Emergency contact management
• Add and update emergency contacts
• Offline SOS communication
• SMS-based emergency alerts
• GPS coordinates in SMS
• Address sharing
• Google Maps location link
• Emergency type information
• Multiple emergency contacts

Working of the Module:

When an SOS is triggered, the application checks the available network connection. If internet connectivity is unavailable, the application prepares an emergency SMS containing the user's emergency information and current location.

The SMS is sent to the configured emergency contact. The recipient can use the shared coordinates or map link to identify the user's location.

MODULE 7 — VOLUNTEER LOCATION & NAVIGATION

The Volunteer Location and Navigation Module helps coordinate movement between the volunteer and the person requesting emergency assistance. After an SOS request is accepted, the system can track the volunteer's location and display it to the user.

The module uses GPS and map services to provide the current position of both parties and support route visualization.

Main Features:

• Volunteer GPS tracking
• Real-time volunteer location
• User location display
• Distance calculation
• Route visualization
• Map-based navigation
• Volunteer movement tracking
• Arrival status
• Location updates

Working of the Module:

After accepting an SOS request, the volunteer's current GPS coordinates are stored in Firebase Firestore. The user's application listens for location updates and displays the volunteer marker on the map.

The system can calculate or display the route between the user and volunteer. As the volunteer moves, updated coordinates can be reflected on the user's map.

MODULE 8 — WEATHER & DISASTER INFORMATION

The Weather and Disaster Information Module provides environmental and disaster-related information that can help users understand current conditions. Weather information can be particularly useful during floods, storms, heavy rainfall, and other weather-related emergencies.

The module can display current weather conditions and relevant alerts based on the user's location.

Main Features:

• Current weather information
• Temperature
• Humidity
• Rainfall information
• Wind information
• Weather forecast
• Location-based weather

• Weather alerts
• Disaster-related warnings
• Emergency information updates

Working of the Module:

The user's location is obtained through GPS and used to request weather information from the configured weather service. The retrieved information is processed and displayed in the application.

Weather information can also be associated with emergency conditions and displayed as alerts when relevant.

MODULE 9 — EMERGENCY STATUS & NOTIFICATION

The Emergency Status and Notification Module keeps users and volunteers informed about changes in an SOS request. It provides real-time status updates throughout the emergency-response process.

The status can change as the emergency request progresses from creation to volunteer assignment and resolution.

Main Features:

• SOS status tracking
• Pending status
• Accepted status
• Volunteer arriving status
• Resolved status
• Emergency notifications
• Volunteer response notifications
• Real-time Firestore updates
• SOS history
• Emergency activity tracking

Working of the Module:

When an SOS request is created, its status is initially stored as Pending. When a volunteer accepts the request, the status changes to Accepted. As the volunteer travels toward the emergency location, the status can change to Volunteer Arriving.

After the emergency has been handled, the request can be marked as Resolved. Firebase Firestore real-time listeners allow the application to display these changes without requiring continuous manual refreshing.

MODULE 10 — EMERGENCY HISTORY & RECORD MANAGEMENT

The Emergency History and Record Management Module maintains records of the user's previous emergency requests. It allows users to review their SOS activities and understand the status and response associated with previous emergencies. The module stores important information such as emergency type, location, date, status, and volunteer assignment.

Main Features:

- SOS history
- Emergency request records
- Emergency type details
- Location history
- SOS date and time
- Volunteer assignment details
- Emergency status history
- Resolved emergency records
- Emergency record viewing
- Firebase Firestore storage

Working of the Module:

Whenever an SOS request is created, its details are stored in Firebase Firestore. The application retrieves the user's previous SOS records using the authenticated user ID.

The history screen displays emergency information such as date, location, emergency type, assigned volunteer, and final status. This provides users with an organized record of their previous emergency activities.

MODULE 11 — PROFILE & DATABASE MANAGEMENT

The Profile and Database Management Module manages the user's personal information, emergency contacts, preferences, and application data. It connects the user's identity with the information required by the different Disaster Help Center modules.

Users can update their name, phone number, address, emergency contacts, and other profile information. The database component manages SOS requests, volunteer information, location data, and emergency records.

Main Features:

- User profile management
- Personal information management
- Phone number and address management
- Emergency contact management
- Account settings
- User data storage
- SOS request records
- Volunteer assignment records
- Location data management
- Emergency history
- Secure data retrieval and updates

Working of the Module:

The Flutter application sends profile and database requests to Firebase services. Firebase Authentication manages the user's identity, while Firebase Firestore stores profile information, emergency contacts, SOS requests, volunteer assignments, and other application records.

When the user updates profile information, the application validates the data and updates the corresponding Firestore document. Other modules can then retrieve the required information to provide personalized emergency services.